

Project: Poker Game Simulator with Expectiminimax

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Course: CSE440

Section: 01

Group: 7

Week 1 (Feb 10 – Feb 16): Project Planning, Topic Finalization, and Algorithm Research

I finalized the project topic with my teammates: a simplified Texas Hold'em poker simulator with an AI bot using the Expectiminimax algorithm. I read the updated course manual carefully to understand project requirements, GitHub, deadlines, and report formats. I reviewed lecture materials and online resources on minimax and expectiminimax, focusing on how chance nodes are modeled in card games. Based on this understanding, I prepared an initial project plan dividing the system into modules (game engine, hand evaluation, AI search, testing, and documentation) and shared this structure with my teammates.

Week 2 (Feb 17 – Feb 23): System Architecture Design and Poker Rule Specification

I designed the preliminary architecture of the simulator and determined how to separate the core modules such as the game driver, game-state manager, card utilities, hand evaluator, and AI decision component. I also wrote a short specification describing the simplified poker rules our simulator will follow, including number of players, betting structure, available actions, and order of rounds. In addition, I created a basic block diagram showing how data will flow between modules and discussed these design choices with the group to maintain consistency in data representation.

Week 3 (Feb 24 – Mar 1): Core Game Engine Development and Initial Simulator Prototype

During this week I began developing an initial version of the game engine. I implemented the deck structure, card shuffling logic, and mechanisms for dealing hole cards and community cards while ensuring no duplication occurs. I also started constructing the main game loop that manages turn order, pot updates, and transitions between stages such as pre-flop, flop, turn, river, and showdown. A temporary text-based interface was created to run early test rounds so that we could observe the basic flow of the simulator.

Week 4 (Mar 2 – Mar 8): Hand Evaluation Module Development and Scoring Framework

I worked on the first version of the hand evaluation module that identifies standard poker hand categories (such as pair, two pair, straight, flush, etc.) using the seven available cards. I experimented with different ways of assigning numeric scores to hands so that the evaluation values can later be used within the Expectiminimax search. At this stage the evaluator is functional but still requires refinement and extensive testing to ensure correct rankings and consistent behavior across different edge cases.

Week 5 (Mar 9 – Mar 15): Expectiminimax Integration and AI Decision Framework Implementation

I began integrating the AI component based on the Expectiminimax algorithm. I defined a state representation that includes the current player, public and private cards, pot size, chip counts, and simplified betting history. I also started building the recursive Expectiminimax structure with MAX nodes (AI decisions), MIN nodes (opponent responses), and CHANCE nodes (card draws). The integration is still in progress, and the AI currently operates in a preliminary testing mode while we continue improving search depth control and evaluation accuracy.

Current Status and Next Steps

So far I have led the overall planning of the project, designed the system architecture, implemented an early version of the game engine, and developed initial components for hand evaluation and Expectiminimax integration. The project is currently in the development and testing stage. Over the next two to three weeks I plan to complete the AI integration, refine the evaluation function, conduct systematic testing against baseline strategies, and coordinate the preparation of the final report, presentation slides, and the organized GitHub repository.
